

a Donor-aware partitioning of PBMC CITE-seq atlas

Public PBMC CITE-seq atlas
Hao et al., 2021
8 donors

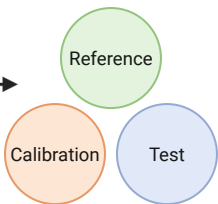


Reference donors
P3, P4, P5, P8

Calibration donors
P2, P7

Test donors
P1, P6

No donor overlap

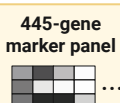


b Frozen PBMC signature and benchmark mixtures

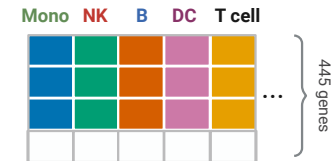
Reference donors



DE marker selection



Five-cell-type signature matrix



Calibration/Test donors



Mixture sampling

Pseudo-bulk mixtures

Dirichlet(1.0)
500 cells per mixture

Known cell fractions used for calibration/test evaluation

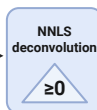
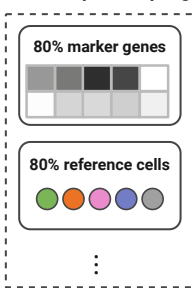


500 calibration + 500 test mixtures

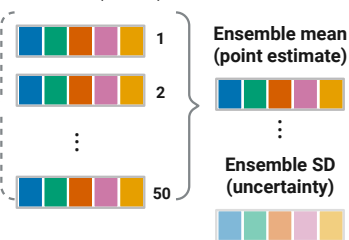
c Bootstrap NNLS ensemble for uncertainty quantification



Bootstrap resampling

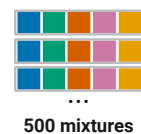


Ensemble ($B = 50$)

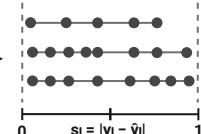


d Split conformal calibration

Calibration set
(with known fractions)



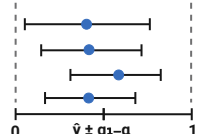
Nonconformity scores
(absolute error)



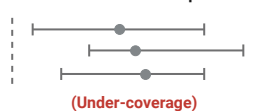
Conformal quantile



Calibrated prediction interval (per cell type)



Raw ensemble 5th-95th percentile



Conformal interval ($q_{1-\alpha}$)



Empirically calibrated marginal intervals